ITECH3208 Capstone · Federation University Australia

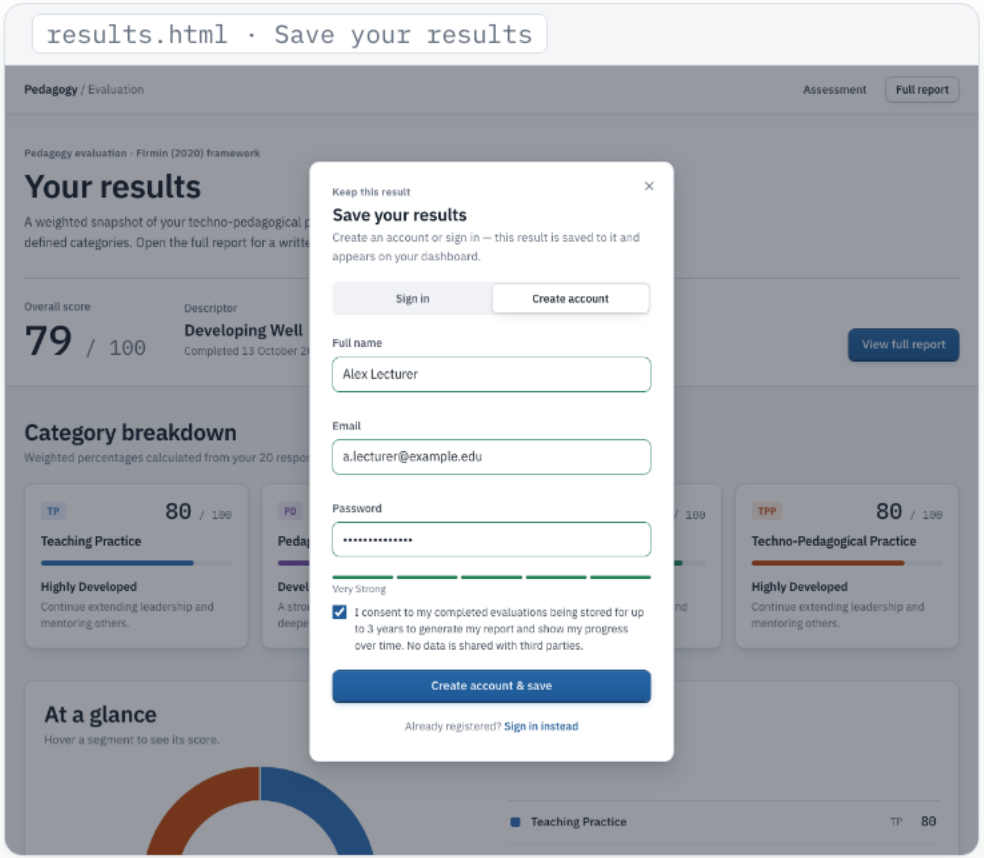
Pedagogy Evaluation Platform

Sprint 5 review

Anonymous start, saved result · Email verification code · Question bank for administrators

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Client: Dr Selena (Sally) Firmin



Sprint 5 in numbers

Both requirements shipped — plus the one that arrived mid-sprint.

3 / 3

requirement areas delivered and live

15

acceptance criteria · 12 verified by the team, 3 at the tester's live run

7

pull requests merged · 6 production deployments, 12–15 October

+1,695

lines added · −379 removed · 28 files

113

automated database checks, all passing · 73 migration, 40 stress

3

database migrations applied to production

24 h

window in which a stored anonymous result can be saved to an account

0

new credentials — the result's random session id is the only key

What the client asked for

Two requirements, one mid-sprint request.

Ref	Client's words	Shown on	Status
R1	Anonymous user registration and result saving — start anonymously; at the end, offer to create an account; the completed result is saved to that account and can be viewed later.	Slides 4–6, 8	LIVE
R2	Email verification code — when a user registers, a code is sent to the email address; the account is verified by entering that code.	Slide 7	LIVE
QB	Question bank — see all questions in the admin panel and edit them.	Slides 10–11	LIVE
+	Forgot password — added 13 October after the review build.	Slide 12	LIVE
+	Archive instead of delete — client rule, added 14 October.	Slide 13	LIVE
+	Time on task — client request, added 15 October.	Slide 14	LIVE

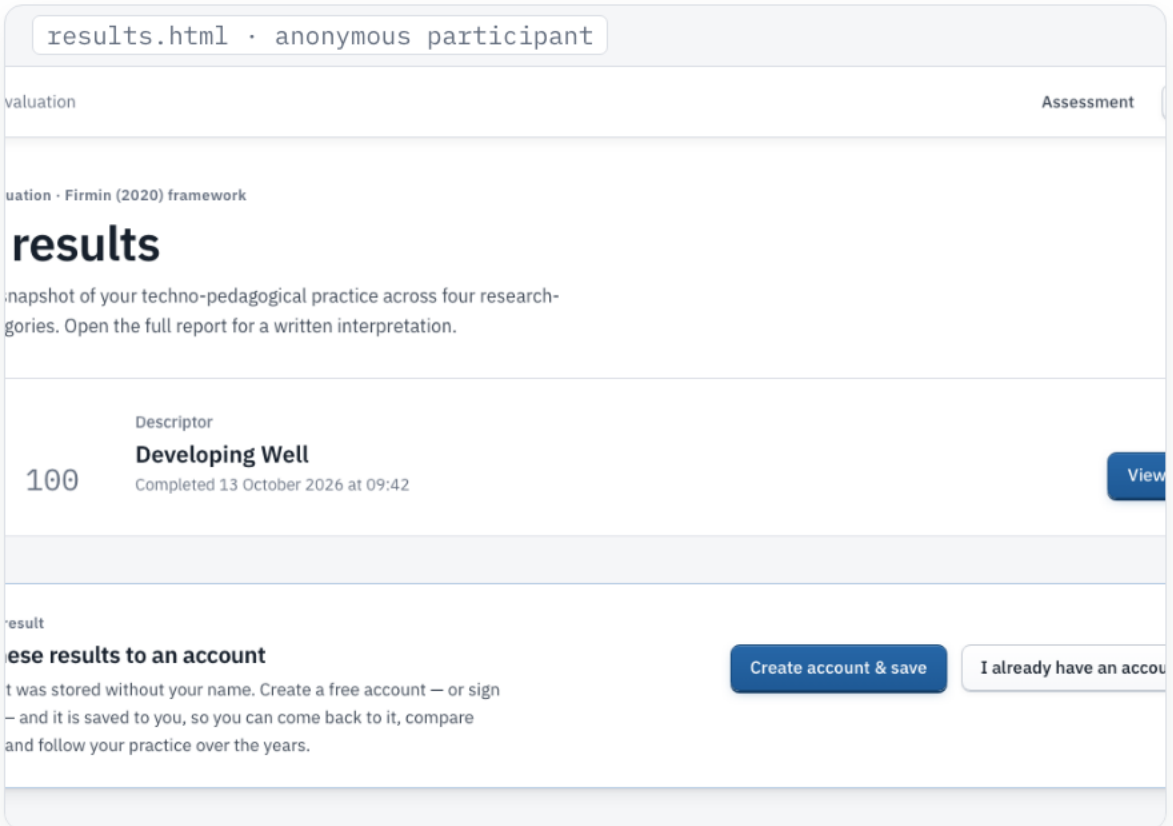
HOW WE READ R1

The result the participant has just seen becomes the account's result. No retake, no re-entry — the offer is made on the results page, after submission.

Requirement 1 · Anonymous start, saved result

Start with nothing to lose. Keep what you've seen.

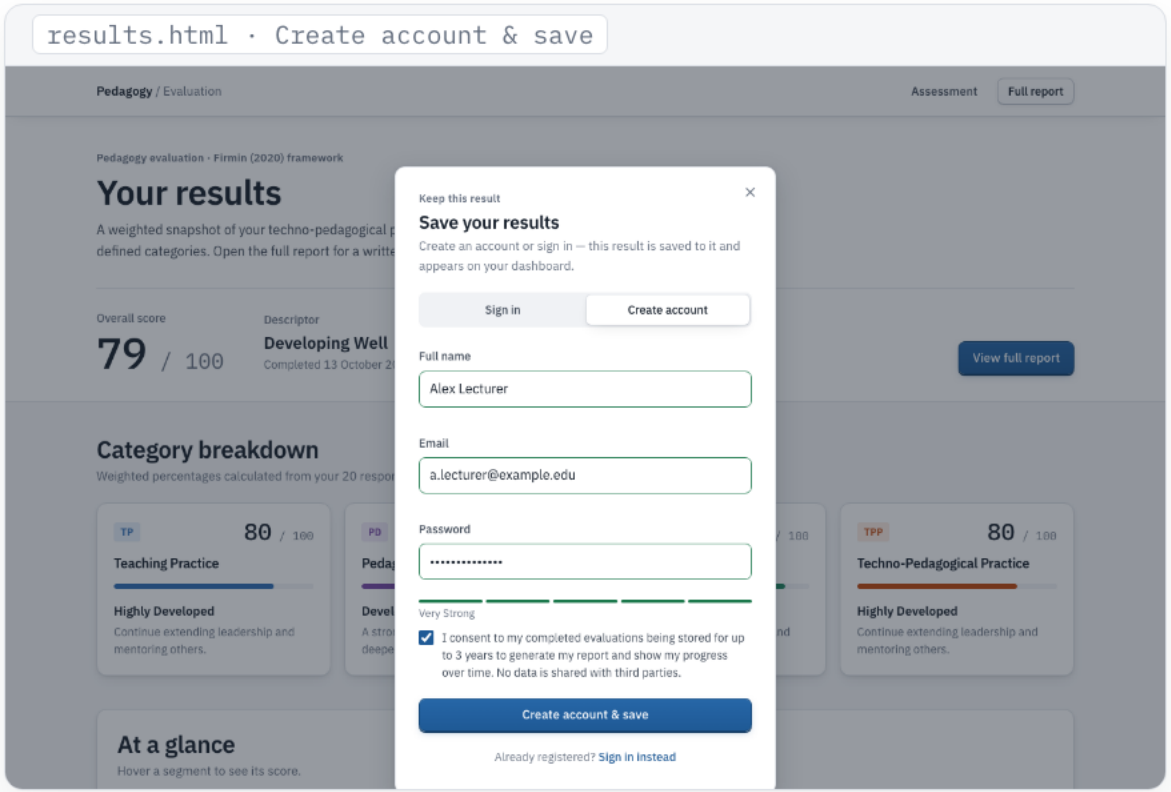
- **Nothing changes until the result is on screen.** No account, nothing stored while answering, one write on submit.
- **The results page makes the offer** — "Save these results to an account" — only when the result actually reached the server.
- **Two ways in:** create an account, or sign in to one you already have.
- **Close the modal and nothing happens.** Still anonymous, card unchanged.



Requirement 1 · Save your results

The same sign-in, told the truth about what it does.

- **Title "Save your results",** not "Start the assessment".
- **No Anonymous tab** — they are already anonymous.
- **Buttons say what happens:** "Create account & save", "Sign in & save results", "Verify & save results".
- **Refresh mid-way?** The code panel comes back, still in save mode.

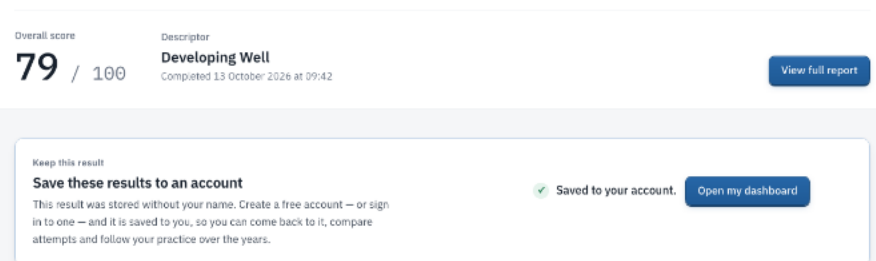


Requirement 1 · Saved to the account

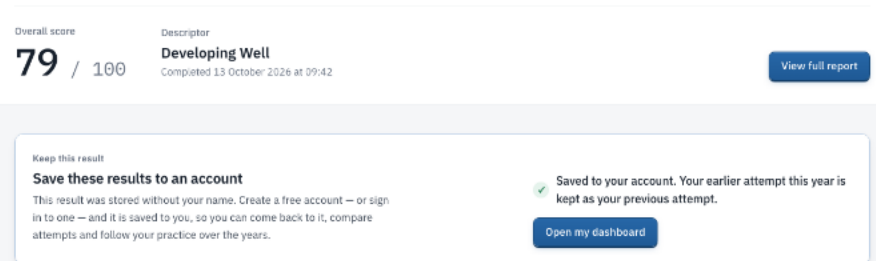
Scores stay on screen. The result moves to the account.

- **"Saved to your account"** with a link straight to the dashboard.
- **Already have an attempt this year?** It is kept as your previous attempt — the same rule as a normal second submission.
- **If the save fails,** the card says why and the account still works; the next evaluation saves automatically.
- **Available while the results tab is open,** and on the server for 24 hours.

first attempt this year

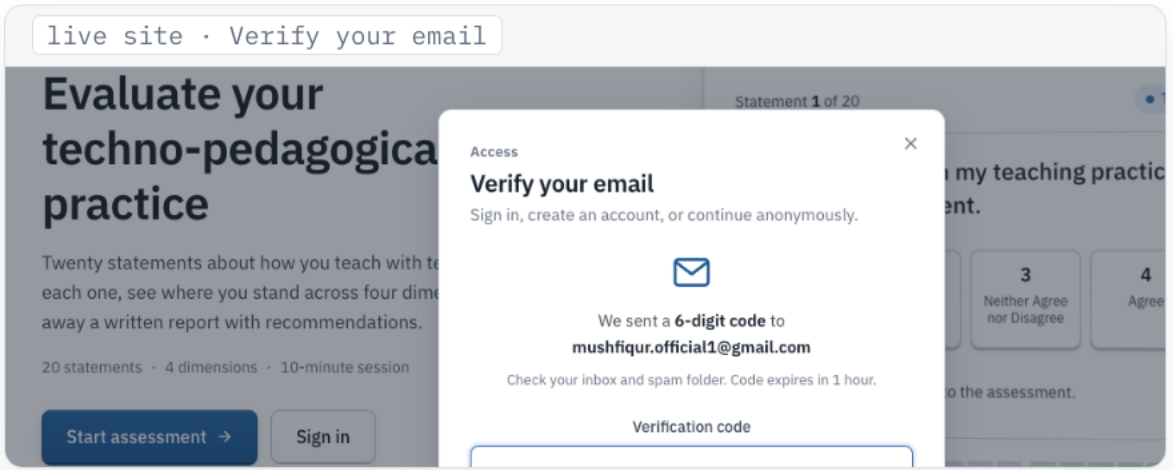


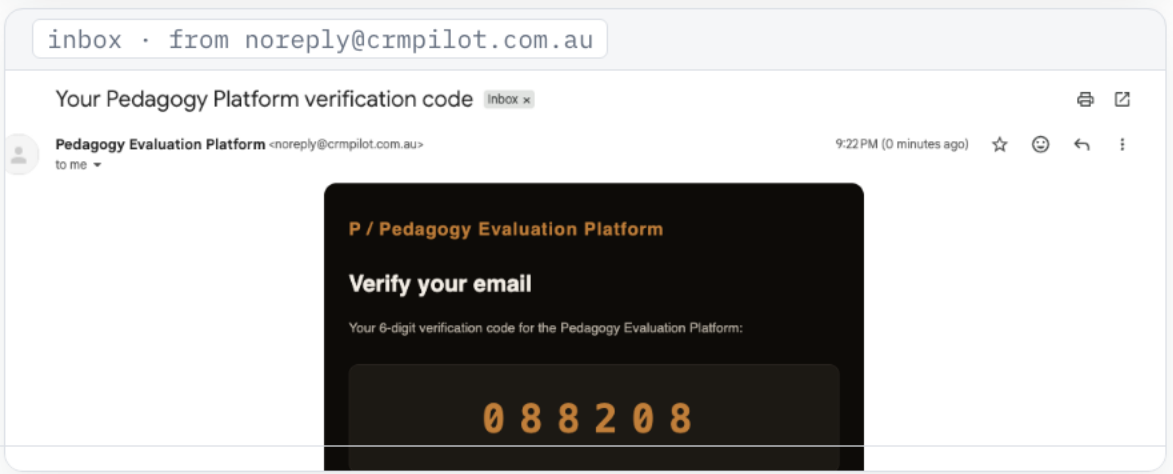
earlier attempt kept as previous



A 6-digit code, not a link.

- **Register → code emailed** to the address given; the account is unusable until it is entered.
- **Wrong or expired code is refused;** codes last one hour; "Resend code" sends a new one and only the newest works.
- **Try to sign in before verifying?** Back to the code screen with a fresh code.
- **Works from the home page and from the results page** — same code, same panel.
- **Configured 13 October:** sent from noreply@crmpilot.com.au; a live registration got its code within a minute.





Under the hood

The rules live in the database, not the page.

```
claim_assessment(p_session_id)

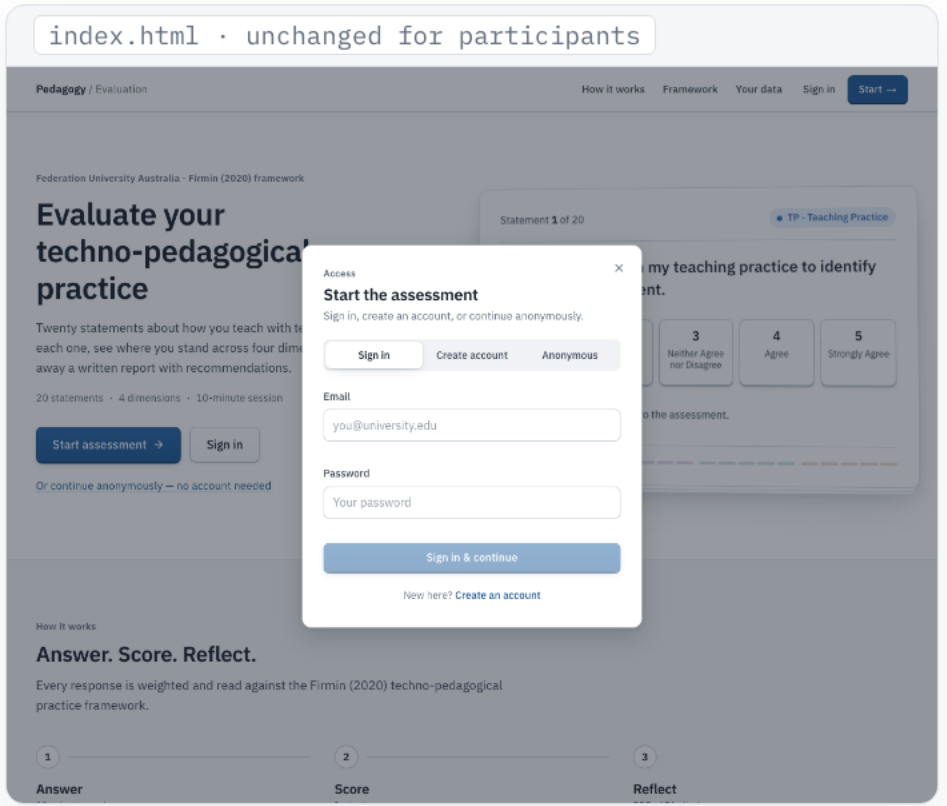
 1  signed in, not an anonymous token   → refused
 2  completed, still has no owner       →
CLAIM_UNAVAILABLE
    row locked — two claims cannot race
 3  completed under 24 hours ago        →
CLAIM_EXPIRED
 4  owner = caller, email = account address
 5  keep this result + the newest earlier
    same-year attempt; remove older ones
    (the Sprint 4 two-attempts rule)
 6  return { year, previous_id, removed }
```

- **One credential: the session id.** A random UUID returned only to the browser that submitted. Nobody can list or guess unclaimed results.
- **The anonymous role cannot even call it.** No grant, and a second check inside.
- **Once only.** A claimed result has an owner; the function refuses it from then on.
- **Nothing new to protect.** No new table, no token, no email link — Sprint 4's rules, reused.

Under the hood

One sign-in modal, usable from any page.

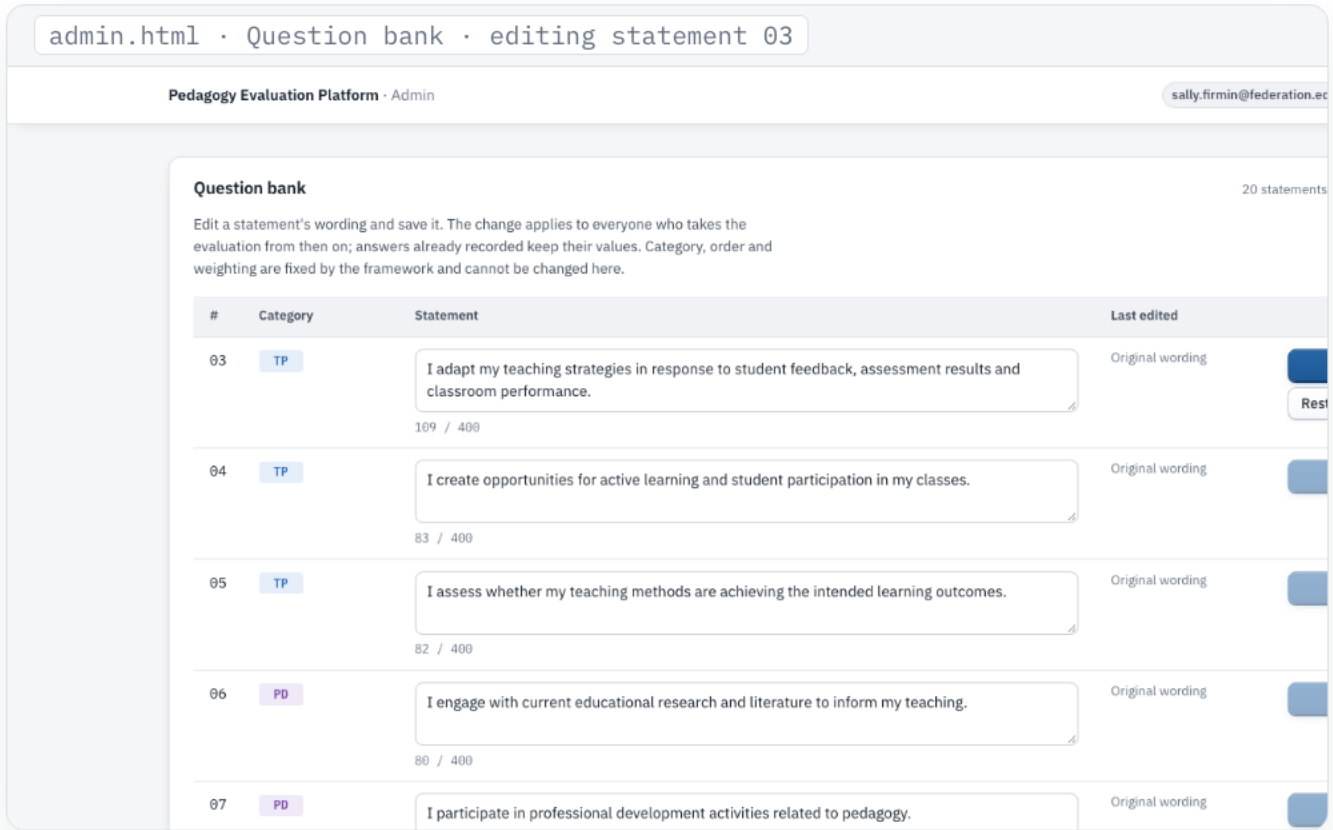
- **The modal is rendered by auth.js,** not copied into each page. The home page lost 100 lines of duplicate markup.
- **Intents.** Opened to *start* (go to the questionnaire) or to *claim* (stay here and save). The intent sets the copy and hides the Anonymous tab.
- **A single event, ped:signed-in.** After sign-in or verification the page decides what happens next; auth.js only redirects if nobody takes over.
- **identity.adopt().** Becoming an account holder no longer wipes the result on the device — that was the first bug we found.



Question bank · requested 12 October, live 12 October

The client owns the wording of the instrument.

- **All twenty statements in the admin dashboard,** with category, character counter and who last edited it.
- **Save, or Ctrl+Enter.** Restore original brings back the shipped wording.
- **Participants read the database first** — home page, questionnaire, review screen, admin responses — and fall back to the shipped copy if it cannot be reached.
- **Shipped read-only first,** with a notice, until the migration was applied.



Question bank · guardrails

Only the words can change. The scoring model cannot.

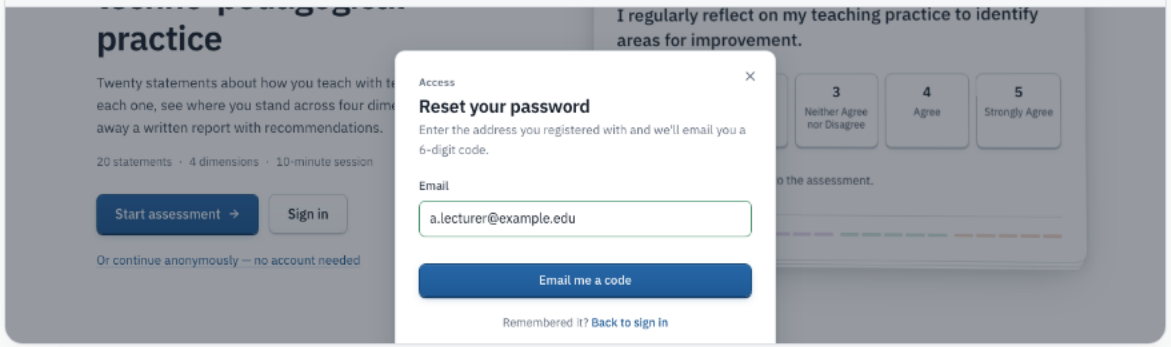
Rule	Enforced by the database
Only administrators	Row-level security policy for the two administrator addresses. Anyone else's update affects zero rows — verified live.
Only the wording	A trigger refuses any change to code, category, weight, order or active flag: "Only the wording of a question can be edited". Five per category, equal weights — fixed.
A real sentence	Trimmed, 10 to 400 characters, or the save is refused.
Every edit stamped	When, and by whom — shown in the panel, counted in the header.
Past answers untouched	Responses store the value and the statement code. An edit changes what future participants read, nothing already recorded.

Added after the review build · live 13 October

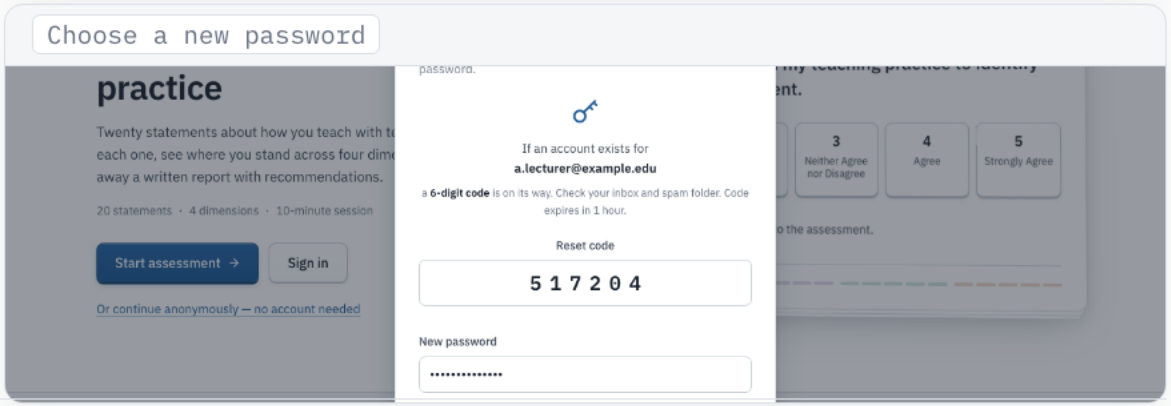
Forgot your password? A code, a new password, signed in.

- **"Forgot password?"** under the sign-in password field, on any page that opens the modal.
- **Email → 6-digit code,** same sender as registration. Same reply for unknown addresses, so nobody can probe who is registered.
- **Code + new password → signed in.** From the results page, the result is saved to the account too.
- **Live since 13 October:** built, tested on its preview, reset template set, merged and deployed as version 2.4.

Reset your password



Choose a new password

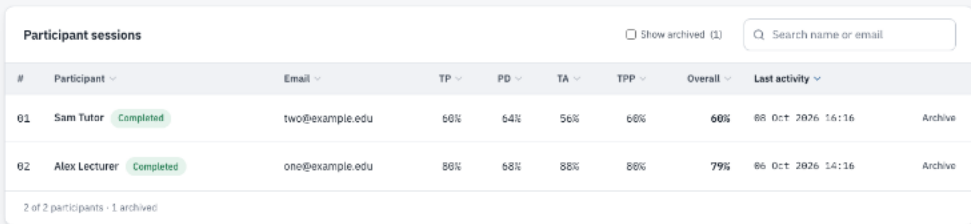


Added 14 October · client rule · live

Administrators archive. They never delete.

- **Delete is gone** from the dashboard, and the administrator delete policies are gone from the database. An admin delete now affects zero rows, whatever sends it.
- **Archive** hides a participant from the table, stats, charts and export. Nothing is removed.
- **Show archived** reveals them greyed, badged with when and by whom. **Restore** brings them back.
- **Guarded:** only administrators, never on their own account, and a participant cannot change their own state. 20 database checks; live as version 2.5.

`admin.html · Archive in place of Delete`



`Show archived · Restore`

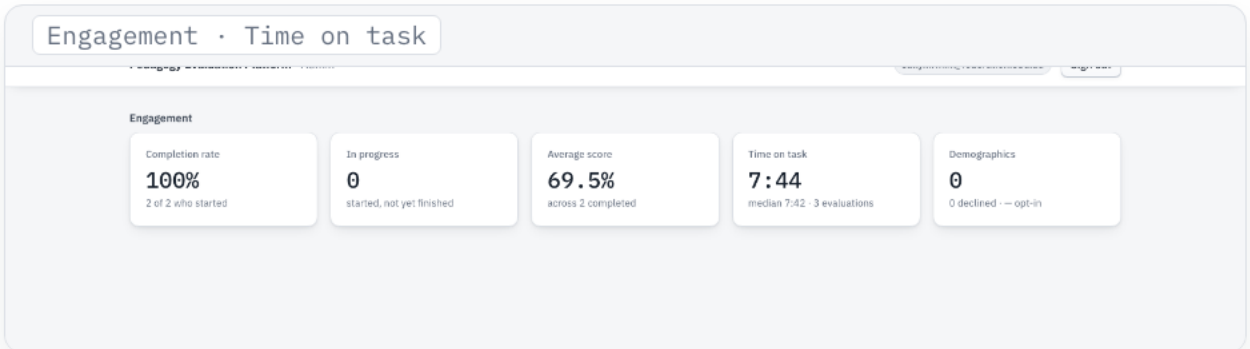


Added 15 October · client request · live

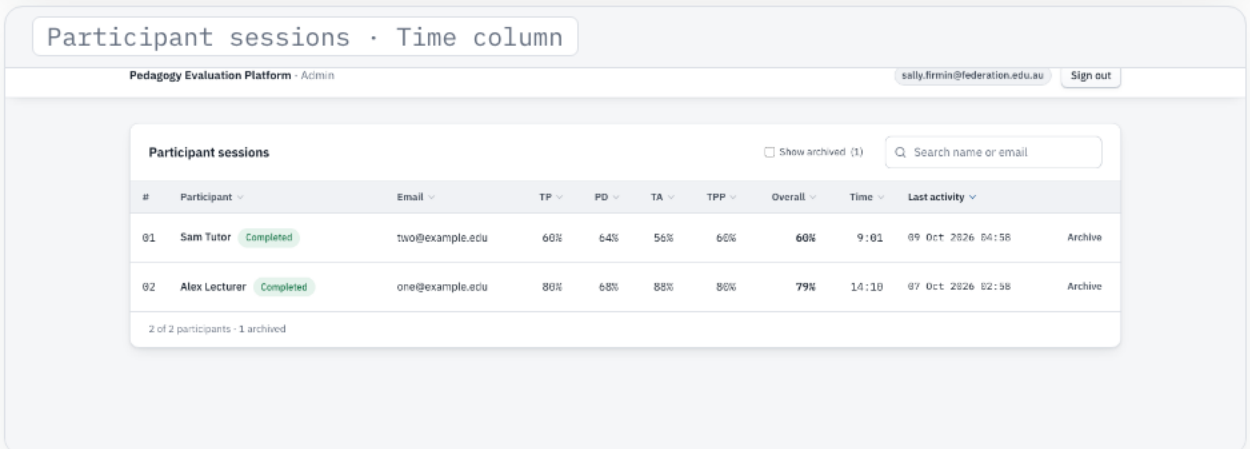
Time on task, from data we already had.

- **No new tracking.** Each evaluation already stores its start and submit times; time spent is the difference.
- **Time column** per participant: total across attempts, sortable, latest on hover.
- **Time on task card:** average and median per evaluation; archived excluded.
- **Profile and CSV too.** Old rows capped at 11:00 so they can't skew it.

Engagement · Time on task



Participant sessions · Time column



Live demo

Let's use it.

1. **Home** — continue anonymously, answer the twenty statements, submit
2. **Results** — "Create account & save" → enter the emailed code → "Saved to your account"
3. **Dashboard** — the same scores, now the account's active result
4. **Admin** — edit one statement's wording, then show it on the home page

`pedagogy-eval.vercel.app`

How we tested

Three levels, same as Sprint 4.

113

Database checks

- Real Postgres engine, full schema chain, mocked sign-in
- **32** for the claim: once only; anonymous, no-user, cross-user refused; 24-hour window; two-per-year trim; grants
- **21** for the question bank: length, each locked column, trim, stamp, admin vs non-admin
- **20** for archive: admin only, self-archive refused, delete policies gone, admin delete = 0 rows
- **40** stress: 300 submissions, bad inputs, claim races, 400-char edits, 150 archives

Browser

End to end

- Card only with a server result and an anonymous identity
- Claim modal copy, hidden tab, close behaviours, refresh during verification
- Saved, previous-kept and failure states — scores unchanged
- Question bank: counter, validation, restore, read-only state
- Forgot password and archive flows with the service stubbed
- No console errors, assets at 2.5

Live

Read-only probes · 13–14 Oct

- Site serves 2.5; claim, archive and question-bank functions present, anonymous role refused
- Reset function gone; email confirmation required
- **A code arrived** from noreply@crmpilot.com.au — live registration
- **Stress:** 11 hostile requests + a 30-call burst, all refused; phone walk-through, nothing sent
- **Still to run:** live claim, admin save, password reset, archive — tester plans

Acceptance criteria

Fifteen criteria. Twelve signed off; three at the live run.

#	Criterion	Evidence	Result
1	Start the evaluation anonymously	Sprint 4 live run	MET
2	Offered an account at the end	Slide 4 · browser	MET
3	Existing account can sign in and keep the result	Slide 5 · browser	MET
4	Result saved to the account	DB tests · live probe	MET
5	Result viewable later on the dashboard	DB tests · live run A9–A10	LIVE RUN
6	Scores stay on screen, no retake	Slide 6 · browser	MET
7	Only the submitting browser, once, within 24 h, attempt rules applied	DB tests · live probe	MET
8	Registering sends a 6-digit code	Slide 7 · live registration	MET
9	Verified only by the correct code	Browser · live run A7–A8, C3–C4	LIVE RUN
10	Admins see all twenty statements	Slide 10 · live probe	MET
11	Admin edit reaches participants	DB tests · live run B4–C2	LIVE RUN
12	Only wording; stamped; past answers unaffected	DB tests · live probe	MET
13	Added: a forgotten password reset with an emailed code	Slide 12 · browser · live	MET
14	Added: administrators archive a participant, never delete one	Slide 13 · DB tests · live	MET
15	Added: how long each participant spent	Slide 14 · browser · live	MET

What we caught

Found in the build, fixed before anyone noticed.

Found	Impact	Fix
Creating the account wiped the result from the device	The claim would have deleted the id it needed	New identity.adopt() — switch identity, keep the result
Modal could only open from the home page	No way to sign in on the results page	Rendered by auth.js; one source, any page
Question bank merged into the wrong branch	Looked merged, live site still 2.2	Caught in minutes by the live version; third pull request to main
Test engine crashed on the migration's refusals	No checks would run on PGlite 0.3	Pinned to 0.2.17; all 53 pass
Completion screen said nothing about saving	The offer on the next page came as a surprise	"On the results page you can save them to an account"

Handover

Handover plan.

Item	Why it matters	Status
Verification-code sender	Custom SMTP on a verified domain, noreply@crmpilot.com.au — a code arrived in a live test	DONE 13 OCT
Tester live run	Real claim, real admin edit — closes criteria 5, 9 and 11	SCHEDULED
Archive instead of delete	Migration applied, merged and live — version 2.5	DONE 14 OCT
Time on task	Built from the timestamps already stored; live — version 2.6	DONE 15 OCT
Forgot password	Built, tested, template set, merged and live — version 2.4	DONE 13 OCT
Results-email sender	Point the results function at the same sender as the codes	HANDOVER
Credential rotation	Keys and passwords in the handover notes change hands cleanly	HANDOVER
Keep-alive and backups	The free tier pauses after a week idle and keeps no backups	HANDOVER
Ownership and handover guide	University accounts; setup, seven-file migration order, recovery	HANDOVER

What's next

Five actions, then the platform leaves the team.

1. **Live run** — Merajun completes test plans A–B and B, and a real password reset
2. **Results email** — point the results function at the same sender as the codes
3. **Rotate and transfer** — every credential, then ownership to university accounts
4. **Keep-alive and export** — daily ping, scheduled CSV export
5. **Handover guide** — so none of this depends on one person

Questions.

Requirements → Albin · Technical → Mushfiqur · Testing → Merajun · Timeline, handover and cost → Shafiullah

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